Robotics 4/M – 2025

Section A: Sensors, Actuators & Image Processing

Tutorial 1: Sensors

Q1.
a. Define what a Sensor is and provide three examples with applications.

b. The Silicon Sensing Orion Accelerometer has the following specifications:

Selectable Range: ±2.5g or ±10g
Resolution: 0.079mg (±2.5g), 0.313mg (±10g)
Non-linearity: <±3mg (±2.5g), <±5mg (±10g)

Explain the significance of each number provided.

c. Given the following specification:

Range: 0-220°C
emf_0: 0mV
emf_{100}: 5.287mV
emf_{220}: 11.436mV

Calculate the non-linearity error at 100°C as a percentage of the full range output, assuming a linear relationship between the emf and temperature over the full range. Work to 3 Decimal Places.

Q2.

Explain what is sensor range, resolution, dead band and sensitivity.

Q3.
A potentiometer is used as an angular position sensor for a rotating joint in a robotic arm. The maximum resistance, R, of the potentiometer is 500Ω. When the joint is at 0° the resistance, R_p, is 100Ω, at 22.5° R_p is 175Ω, at 45° R_p is 250Ω, at 67.5° R_p is 325Ω and at 90° R_p is 400Ω. Work to 3 decimal places.

a. What is the resistance when the joint is at an angle of 60°?

b. With a Vs of 10V and no load, what are the output voltages for 0°, 45° and 90°?

c. With a load of 5KΩ, what are the output voltages for 0°, 45° and 90°?

Noting that $V_O = V_S.(\frac{x}{\frac{R}{R_L}.x.(1-x)+1})$

d. What is the full-scale error, as a percentage, in the output voltages for 0°, 45° and 90° with a load of 5KΩ?

Noting that $error = V_S.\frac{R}{R_L}.(x^2 - x^3)$

Q4.
a. What is the difference between an *Absolute Encoder* and an *Incremental Encoder*?

b. Explain the basic working principle of an optical encoder. A figure can be used.

c. If an encoder has 8 tracks, what is the achievable resolution to 3 decimal places?

d. On a single-track encoder, how many lines (or cutouts) are required to achieve a resolution of 0.1°?

e. Draw a 4 track gray code encoder

Q5.
a. Describe the Time of Flight method for range finders.

b. Describe the Optical Triangulation method for range finding.

c. An Ultrasonic range finder is being used and an echo is detected after 500μs. How far away is the object?

d. With respect to Optical Triangulation, given L at 50cm, α = 80° and an object distance of 0.425m. What is the angle β?

Q6.
a. What will be the change of resistance of an electrical resistance strain gauge with a gauge factor of 2.25 and resistance of 50 Ω if it is subjected to a strain of 0.002? Use Equation 1.

$$\frac{\Delta R}{R} = K.\frac{\Delta L}{L} = K.\varepsilon \tag{1}$$

b. A metal rod has an axial length of 200mm. A strain gauge with a gauge factor of 2.2 and a nominal resistance of 250Ω is attached to the rod to measure axial strain. An axial tensile load is applied to the rod. The single strain gauge is connected into a Wheatstone bridge circuit with a supply voltage of 10V. The output voltage from the bridge changes by 0.02V. What is the new length of the rod? Use Equation 2.

$$V_O = V_S.\left(\frac{R_2}{R_1+R_2} - \frac{R_x+\Delta R}{R_x+R_3+\Delta R}\right) \tag{2}$$

Past Paper Questions

2007 Q1

(e) A robotic rover vehicle uses an optical triangulation system for range finding: [7]

(i) Sketch this system indicating the Transmitter angle (α), Receiver angle (β) and the distance between transmitter and receiver (L).

(ii) Derive from first principles an expression for the object distance in terms of α, β and L.

(iii) Given that a 1000 line incremental encoder is used to measure transmitter angle, determine the resolution in distance at a range of 2m, for $\beta = 85°$ and L = 50cm.